Instruction set

This 16-bit ISA controls an 8-bit, five-stage washing-machine processor. All eight registers, including R0, are writable. Instruction addresses are word addresses from 00 to FF hex.

Three instruction formats

Format	Bits 15 to 12	Bits 11 to 9	Bits 8 to 6	Bits 5 to 0
R	Opcode	Rd	Rs	Ignored; assembler writes 0
I	Opcode	Register	Bit 8 ignored; bits 7 to 0 form imm8 / target8	Part of imm8 / target8
C	Opcode	Ignored	JMP uses bits 7 to 0 as target; other low bits ignored	Part of JMP target; otherwise ignored

R format: MOV, ADD, SUB, AND, LD, ST and CMPGEU. I format: LDI, ADDI, BEQZ and BNEZ. C format: NOP, JMP and HALT. Unused bits are ignored by hardware (the assembler writes zero). The branch-test register is in bits [11:9], even though it is named Rs in branch assembly.

Hex opcode	Instruction	Effect
0	NOP	No architectural write; next PC
1	LDI Rd, imm8	Rd = imm8, unchanged bit pattern
2	MOV Rd, Rs	Rd = Rs
3	ADD Rd, Rs	Rd = (Rd + Rs) modulo 256
4	SUB Rd, Rs	Rd = (Rd - Rs) modulo 256
5	AND Rd, Rs	Rd = Rd bitwise AND Rs
6	LD Rd, [Rs]	Rd = DATA[Rs]
7	ST Rd, [Rs]	DATA[Rs] = Rd
8	ADDI Rd, imm8	Rd = (Rd + imm8) modulo 256
9	BEQZ Rs, target	If Rs = 0, redirect PC to target
A	BNEZ Rs, target	If Rs is nonzero, redirect PC to target
B	JMP target	Always redirect PC to target
C	HALT	Stop in EX; kill younger entries; drain older entries
D	CMPGEU Rd, Rs	Rd = 1 if unsigned old Rd >= unsigned Rs, else 0
E and F	Unassigned	Deterministic NOP behavior; no writes or control transfer

Encoding examples and control flow

Format	Example	Machine word	Calculation
R	ADD R1, R2	3280	(3 << 12) | (1 << 9) | (2 << 6)
I	LDI R1, 100	1264	(1 << 12) | (1 << 9) | 64 hex
I	BEQZ R3, 80 hex	9680	(9 << 12) | (3 << 9) | 80 hex
C	JMP 20 hex	B020	(B << 12) | 20 hex
C	HALT	C000	Opcode C (all unused bits zero)

PC starts at 00. Normal execution increments PC modulo 256. All branch and jump targets are absolute 8-bit instruction-word addresses. There are no delay slots. A taken EX-stage control transfer invalidates both younger entries. A valid HALT stops fetch until reset, while older MEM and WB operations finish. A flushed HALT has no effect.

Immediate arithmetic uses the raw 8-bit pattern. For example, FF hex adds 255 modulo 256, which has the same result as subtracting one. Loads and stores use the full unsigned 8-bit address held in Rs where no immediate offset is added.

Custom unsigned comparison

CMPGEU uses the existing gate-level unsigned comparator and selects a zero-extended 0/1 result in EX. It reads the old destination and Rs, then overwrites Rd. Its result follows the normal EX/MEM and MEM/WB forwarding paths. A preceding load into either source needs one bubble, just as for ADD.

During filling, CMPGEU compares water against the target, so both equality and overshoot finish filling. During draining, the operand order is reversed: threshold >= water means water <= threshold. This avoids an equality-only loop that could hang if the sensor skips the exact target. It also avoids signed-comparison errors at 128 through 255.

Source use and control decoding

Read old Rd	Read Rs or branch register	No register source
ADD, SUB, AND, ST, ADDI, CMPGEU	MOV, ADD, SUB, AND, LD, ST, CMPGEU; BEQZ/BNEZ use their bits 11:9 register	NOP, LDI, JMP, HALT, E and F

The hazard unit qualifies comparisons with these use bits and stage valid bits. Immediate bits that happen to resemble a register number cannot cause a false load-use stall. R0 receives the same forwarding, hazard and write treatment as R1 through R7.